FinStack™: Statistical Models & Technical Application Guide

Powered by MURSHIDI Ai

Executive Summary: The Structural Intelligence of FinStack

This technical note provides a comprehensive mapping of the statistical models and technical analysis techniques embedded within the **Global Financial Solutions FinStack™** platform. As a strategic advisor, I have structured this guide to move beyond surface-level features, revealing the underlying patterns of risk and value that the system enables for institutional banking.

1. Probability of Default (PD) Models

The core of FinStack's risk grading intelligence lies in its integration with Global Financial Solutions' global PD engines, specifically tailored for different entity types.

1.1 RiskCalc™ for Private Firms

- **Technique:** Multivariate probit regression and structural modeling.
- **Application:** Used for rating private corporate entities where market data is unavailable. It converts financial ratios (leverage, liquidity, profitability) into an **Expected Default Frequency (EDF™)**.
- **Use Case:** Critical for SME and middle-market corporate lending where financial spreading is the primary data source [1].

1.2 CreditEdge™ for Public Firms

- **Technique:** Merton structural model (contingent claims analysis).

- **Application:** Treats the firm’s equity as a call option on its assets. It uses market equity volatility and total liabilities to calculate the **Distance-to-Default (DtD)** and subsequent EDF.
- **Use Case:** Real-time monitoring of listed corporate borrowers and large-cap institutional exposures [1].

2. Loss Given Default (LGD) & Recovery Modeling

FinStack utilizes a granular approach to loss estimation, focusing on the structural seniority and collateralization of facilities.

2.1 Structure Lite with LGD

- **Technique:** Cash flow waterfall and recovery-by-asset-class modeling.
- **Application:** Aggregates risk mitigants (collateral, guarantees) and applies haircuts based on asset priority and liquidation costs.
- **Use Case:** Determining the facility-specific LGD for top-level multi-option facilities (MOFs) and standalone products [2].

2.2 Commercial Real Estate (CRE) LGD

- **Technique:** Net Operating Income (NOI) stress testing and Loan-to-Value (LTV) migration.
- **Application:** Specifically models the recovery of real-estate backed loans by analyzing property-specific NOI, capitalization rates, and vacancy stressors.
- **Use Case:** Specialized CRE portfolio management and development finance [2].

3. Risk-Adjusted Return on Capital (RAROC)

The Deal Analytics module serves as the primary engine for pricing and profitability benchmarking.

3.1 Risk Return Analysis

- **Technique:** Economic Capital (EC) allocation and Hurdle Rate benchmarking.

- **Application:** Combines the PD (RiskCalc/CreditEdge), LGD, and Exposure at Default (EAD) to calculate the **Risk-Adjusted Return on Capital**. It measures whether a deal's margin covers the cost of risk and capital.
- **Use Case:** Front-end deal structuring and annual facility reviews to ensure pricing alignment with institutional hurdle rates [3].

4. Portfolio & Aggregation Techniques

FinStack employs advanced graph theory to handle complex institutional hierarchies and limit aggregations.

4.1 Directed Acyclic Graph (DAG) Engine

- **Technique:** Topological sorting and graph-based data aggregation.
- **Application:** The Group Aggregation Calculator (GAC) regards assets, collateral, products, and entities as nodes in a DAG. It ensures that any change in a sub-facility or collateral value correctly propagates through the entire hierarchy.
- **Use Case:** Real-time limit monitoring for complex global conglomerates and cross-collateralized facilities [4].

5. Peer & Industry Benchmarking

The system leverages the Global Financial Solutions' Credit Research Database (CRD) for comparative analysis.

5.1 Peer Analysis & Percentile Mapping

- **Technique:** Cross-sectional industry distribution and percentile ranking.
- **Application:** Maps an entity's financial ratios against a global or regional peer group (e.g., RMA or BankFocus).
- **Use Case:** Identifying "Pricing Outliers" and "Risk Anomalies" by comparing a borrower's performance against sector-level medians [5].

Murshidi AI Interpretation: Solving Patterns, Revealing Value

The Pattern

FinStack provides the **Structural Foundation** for risk assessment, but the **Revealed Value** comes from how these models are orchestrated. The transition from a "Spread-only" mindset to a "Risk-Return" mindset is where the bank captures alpha.

The Structure

The integration of **RiskCalc (PD)**, **Structure Lite (LGD)**, and **RAROC** creates a closed-loop pricing system. By using the **DAG Engine**, the bank ensures that capital is not "trapped" in inefficient sub-limits, but is dynamically allocated to the highest-return nodes.

The Move

Implement the **RAROC 2.0 enhancements** (Sector Beta & WACC) as a strategic overlay to these native FinStack models to move from generic risk grading to high-precision pricing [6].

References

[1] Global Financial Solutions. (2019). *FinStack™ User Guide v5.19.31*, Section 8.7: External Rating Models (p. 160-166). [2] Global Financial Solutions. (2019). *FinStack™ User Guide v5.19.31*, Section 12: Structure Lite & 14: CRE (p. 219, 289). [3] Global Financial Solutions. (2019). *FinStack™ User Guide v5.19.31*, Section 15.2: Risk Return Analysis (p. 355). [4] Global Financial Solutions. (2019). *FinStack™ User Guide v5.19.31*, Section 13.8: Standard Calculation Model (p. 283-286). [5] Global Financial Solutions. (2019). *FinStack™ User Guide v5.19.31*, Section 5.10.5: Peer Comparison Reports (p. 107). [6] Murshidi AI. (2026). *RAROC 2.0 CEO Verdict Report*. (Internal Reference).

Disclaimer: This analysis was prepared with the assistance of Murshidi AI, an artificial intelligence advisory system. All conclusions, recommendations, and strategic *assessments represent analytical outputs and do not constitute financial advice. Independent professional verification is recommended before acting on any recommendation contained herein.*